The Qameah as Clifford Gate: Magic Squares, Quantum Error Correction, and Massive Gauge Fields

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We show that the $\mathrm{GL}(2,\mathbb{Z}_N)$ construction underlying $N \times N$ de la Loubère magic squares is isomorphic to the Clifford group for N-dimensional quantum systems (qu$N$its) when N is an odd prime. A single algebraic parameter $s = 2^{-1} \bmod N = (N{+}1)/2$ governs the entire structure: the Heisenberg–Weyl labeling of entries, the circulant/Hankel decomposition $Q = S + A$, the anticommutation $\{S, A\} = 0$, the Proca equation $B_2^T B_2\, A = N\, A$ on the complete graph K_N, and the nilpotent BRST operators $L_\pm$ with $L_\pm^2 = 0$. The first BRST cohomology $H^1(L_+)$ is one-dimensional for every prime N tested ($3 \le N \le 61$), yielding a quantum code $[[N, 1, (N{+}1)/2]]$ whose distance $d = (N{+}1)/2$ equals s itself. We prove universal Newton sum identities for the Lehmer roots $r_k = 2\cos(2\pi k/N)$: odd power sums equal -4^n (range $p < N$) and even power sums equal $(N\binom{2n}{n} - 4^n)/2$ (range $p < 2N$). For $N = 11$, four independent selection criteria converge uniquely: the Fibonacci–triangular coincidence $\binom{11}{2} = 55 = T_{10} = F_{10}$, the E_8 root count $2(N^2 - 1) = 240$, the perfect code distance $d = 6$, and the Euler characteristic $\chi(K_{11}) = N^2 = |\mathrm{HW}(N)/Z(\mathrm{HW})|$. These results connect magic square theory, discrete gauge theory, and quantum error correction through a single algebraic source with zero free parameters.

I. INTRODUCTION

For odd prime N, the de la Loubère construction of an $N \times N$ magic square is equivalent to a linear map $g \in \mathrm{GL}(2,\mathbb{Z}_N)$ acting on pairs of mutually orthogonal Latin squares (MOLS). Setting the parameter $s = 2^{-1} \bmod N = (N{+}1)/2$ yields a distinguished magic square—the *Qameah* of order N (the $\mathrm{GL}(2,\mathbb{Z}_N)$ magic square with parameter $s = (N{+}1)/2$; a distinguished member of the de la Loubère family satisfying the circulant criterion of Proposition 4)—whose algebraic structure we show to be identical to a single-qu$N$it Clifford gate. The identification is a *structural isomorphism*: the $\mathrm{GL}(2,\mathbb{Z}_N)$ gate acts on Heisenberg–Weyl operator labels by matrix multiplication, and this action equals the conjugation action of the corresponding Clifford unitary[1,2].

From this single datum s, with zero free parameters, we derive five results:

(i) **Clifford gate** (Theorem 10): g defines a Clifford unitary U_g with $\det(g) = s$ (extended Clifford group).

(ii) **Proca eigenvalue** (Theorem 13): $B_2^T B_2\, A = N\, A$ on K_N, a massive spin-1 field with $m^2 = N$.

(iii) **BRST code** (Theorem 26): $L_\pm^2 = 0$ yields a quantum code $[[N, 1, (N{+}1)/2]]$ with $d = s$.

(iv) $N = 11$ **selection** (Theorem 37): four independent criteria converge uniquely at $N = 11$.

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(v) **Newton sums** (Theorem 38): universal identities for Lehmer roots $r_k = 2\cos(2\pi k/N)$.

The BRST code shares its homological structure with CSS codes, but whether $[[11, 1, 6]]$ is useful for quantum computation is an open question, not a result of this paper.

Notation. N is an odd prime, $\mathbb{Z}_N = \mathbb{Z}/N\mathbb{Z}$, $\omega = e^{2\pi i/N}$, $s = (N+1)/2$. $\mathrm{HW}(N)$ is the Heisenberg–Weyl group with shift X, clock Z, and Weyl operators $W_{a,b} = X^a Z^b$. K_N is the complete graph with coboundary operators B_1, B_2. S and A are the symmetric and antisymmetric parts of Q_{dev}; $L_\pm = S \pm A/N$.

II. $\mathrm{GL}(2, \mathbb{Z}_N)$ CONSTRUCTION AND WEYL LABELING

A. The $\mathrm{GL}(2, \mathbb{Z}_N)$ de la Loubère Construction

The classical de la Loubère (Siamese) algorithm fills an $N \times N$ grid by stepping up-and-right, wrapping modulo N, and dropping down on collision[13]. For odd N this is equivalent to a linear map over $\mathbb{Z}_N$, which we now make explicit.

Definition 1 ($\mathrm{GL}(2, \mathbb{Z}_N)$ Gate)**.** Let N be an odd prime and define the *construction parameter*

$$s \;=\; \frac{N+1}{2} \;=\; 2^{-1} \bmod N. \tag{1}$$

The *Qameah gate* is the matrix

$$g \;=\; \begin{pmatrix} s & s-1 \\ s & s \end{pmatrix} \;\in\; \mathrm{GL}(2, \mathbb{Z}_N), \tag{2}$$

with offset vector $\mathbf{c} = \left(\frac{N-1}{2},\, 0\right)^T$.

The determinant is $\det(g) = s^2 - s(s-1) = s \equiv 2^{-1} \pmod N$, which is a unit in $\mathbb{Z}_N$ but is *not* equal to 1 for $N \geq 5$. Hence $g \in \mathrm{GL}(2, \mathbb{Z}_N) \setminus \mathrm{SL}(2, \mathbb{Z}_N)$—a fact that will be important when we identify the gate with a Clifford operator in Sec. II G. Numerical verification confirms $\det(g) = s$ for all primes $N = 3, 5, 7, 11, 13$ (verified computationally; code available at `https://github.com/Insightful-Calculations/qhots_v7_calc`).

The order of g in $\mathrm{GL}(2, \mathbb{Z}_N)$ varies with N: it is $8, 4, 24, 40, 12$ for $N = 3, 5, 7, 11, 13$ respectively, with no simple universal formula.

B. Entry Formula and Permutation Property

Given $(i, j) \in \mathbb{Z}_N \times \mathbb{Z}_N$, define the two *Latin-square coordinates*

$$\begin{pmatrix} \ell_1 \\ \ell_2 \end{pmatrix} = g \begin{pmatrix} i \\ j \end{pmatrix} + \mathbf{c} \mod N, \tag{3}$$

so that explicitly

$$\ell_1(i, j) = \left(s\, i + (s-1)\, j + \tfrac{N-1}{2}\right) \bmod N, \tag{4}$$

$$\ell_2(i, j) = s\,(i + j) \bmod N. \tag{5}$$

The magic square entry is

$$Q_{ij} \;=\; N\,\ell_1(i, j) \;+\; \ell_2(i, j) \;+\; 1. \tag{6}$$

Proposition 2 (Permutation Property)**.** *The map $(i, j) \mapsto Q_{ij}$ is a bijection from $\mathbb{Z}_N \times \mathbb{Z}_N$ to $\{1, 2, \ldots, N^2\}$.*

Proof. Since $\det(g) = s \not\equiv 0$, the linear map $(i,j) \mapsto (\ell_1, \ell_2)$ is a bijection on $\mathbb{Z}_N \times \mathbb{Z}_N$. The mixed-radix encoding $(\ell_1, \ell_2) \mapsto N\ell_1 + \ell_2$ is a bijection from $\mathbb{Z}_N \times \mathbb{Z}_N$ to $\{0, 1, \ldots, N^2 - 1\}$. Adding 1 shifts the range to $\{1, \ldots, N^2\}$. □

Moreover, the matrices $L_1 = (\ell_1(i,j))$ and $L_2 = (\ell_2(i,j))$ are individually Latin squares (each row and column contains every element of $\mathbb{Z}_N$ exactly once), and they form a pair of *mutually orthogonal Latin squares* (MOLS): no ordered pair (ℓ_1, ℓ_2) repeats. This MOLS property is the combinatorial reason the magic square has constant row, column, and diagonal sums.

C. Heisenberg–Weyl Group

The shift and clock operators on $\mathbb{C}^N$ satisfy $X\,|k\rangle = |k{+}1\rangle$, $Z\,|k\rangle = \omega^k\,|k\rangle$, and $ZX = \omega\,XZ$. The Heisenberg–Weyl group $\mathrm{HW}(N) = \{\omega^c X^a Z^b : a, b, c \in \mathbb{Z}_N\}$ has order N^3 and center $Z(\mathrm{HW})$ of order N. The N^2 cosets $\mathrm{HW}(N)/Z(\mathrm{HW})$ are indexed by Weyl operators $W_{a,b} = X^a Z^b$.

D. Weyl Operator Labeling

The MOLS decomposition provides a direct dictionary between magic square entries and Weyl operators.

Definition 3 (Weyl Label)**.** To each position $(i,j) \in \mathbb{Z}_N \times \mathbb{Z}_N$ in the Qameah, assign the Weyl operator

$$W(i,j) \;=\; X^{\ell_1(i,j)}\, Z^{\ell_2(i,j)}. \tag{7}$$

Since (ℓ_1, ℓ_2) ranges over all of $\mathbb{Z}_N \times \mathbb{Z}_N$ as (i,j) does (Proposition 2), this map is a bijection: the N^2 entries of the Qameah label all N^2 elements of $\mathrm{HW}(N)/Z(\mathrm{HW})$, with the identity operator $W_{0,0} = I$ occupying exactly one cell.

The Latin-square property ensures each row contains every X-exponent and every Z-exponent exactly once (verified computationally for $N = 3, 5, 7, 11, 13$).

E. S/A Decomposition and Circulant Structure

Let $Q_{\mathrm{dev}} = Q - \bar{Q}$ denote the deviated magic square (entries centered to have zero mean). The symmetric and antisymmetric parts are

$$S = \tfrac{1}{2}\big(Q_{\mathrm{dev}} + Q_{\mathrm{dev}}^T\big), \qquad A = \tfrac{1}{2}\big(Q_{\mathrm{dev}} - Q_{\mathrm{dev}}^T\big). \tag{8}$$

For the $\mathrm{GL}(2, \mathbb{Z}_N)$ construction, the coordinate $\ell_2 = s(i+j)$ is symmetric in (i,j), so L_2 contributes only to S; while $\ell_1 = s\,i + (s{-}1)j + c$ has coefficient sum $s + (s{-}1) = N \equiv 0 \pmod N$, so L_1 depends only on the displacement $d = j - i$ and generates a *circulant* antisymmetric matrix A.

Proposition 4 (Circulant Criterion)**.** *For a* $\mathrm{GL}(2, \mathbb{Z}_N)$ *magic square with Latin-square coordinates* $\ell_k(i,j) = a_k i + b_k j + c_k$*, the antisymmetric part* A *is circulant if and only if* $a_k + b_k \equiv 0 \pmod N$ *for the antisymmetric contributor.*

Proof. If $a + b \equiv 0$, then $\ell(i, i{+}d) = (bd + c) \bmod N$ depends only on d, so $A_{ij} = A_{i+1,j+1}$ (circulant). Conversely, if $a + b \not\equiv 0$, the entry $\ell(i, i{+}d)$ depends on i through $(a{+}b)i$, breaking circulant symmetry. □

For the Qameah gate (2), the L_1 coefficients $(a,b) = (s, s{-}1)$ satisfy $a + b = N \equiv 0$, so the criterion is met. The resulting decomposition is:

- S is a *Hankel matrix* (constant anti-diagonals), determined by the function $h(u) = s(2u \bmod N)$ reduced to centered form.
- A is an *antisymmetric circulant*, determined by $a(d) = N\,f(d)$ where $f(d) = (sd + c) \bmod N$ is the displacement function.
- The two are related by the structural identity $h(u) = a(u{+}1)/N$ (verified computationally), which links the Hankel and circulant generators through a single shift.

Remark 5. The standard de la Loubère algorithm uses the gate $\left(\begin{smallmatrix}1 & 1\\ 1 & 2\end{smallmatrix}\right)$ with $a + b = 3 \not\equiv 0 \pmod N$ for $N \geq 5$. It produces a valid magic square but with *non-circulant* A, and all subsequent spectral and gauge-theoretic results fail. The $\mathrm{GL}(2, \mathbb{Z}_N)$ Qameah is the unique de la Loubère variant satisfying the circulant criterion.

Proposition 6 (Extremality of the Canonical Construction)**.** *Among all* $\mathrm{GL}(2, \mathbb{Z}_N)$ *de la Loubère constructions with fixed offset* $c = (N-1)/2$*, the canonical choice* $s = (N+1)/2$ *uniquely maximizes* $\|A\|_F^2$*. Precisely,* $\|A_{\mathrm{canon}}\|_F^2 = 2\,\|A_{\mathrm{other}}\|_F^2$ *for every non-canonical* s *coprime to* N*.*

Proof. Write $\delta(i,d) = \ell_1(i, i{+}d) - \ell_1(i{+}d, i)$, the integer lift of $-d \bmod N$. Since the residue is $N{-}d$, each $\delta(i,d)$ is either $(N{-}d)$ or $(-d)$ (*binary lift*).

Canonical. For $s = (N{+}1)/2$ the coefficient $2s{-}1 = N \equiv 0$ kills the i-dependence, so $\delta(i,d)$ depends only on d and A is circulant. The canonical double sum is $S_{\mathrm{c}} = N \sum_{d=1}^{N-1} \delta_d^2 = N^2(N^2{-}1)/3$.

Non-canonical. For $s \neq (N{+}1)/2$, the multiplier $\alpha = (2s{-}1)$ is nonzero and coprime to N, so $\ell_1(i{+}d, i)$ ranges uniformly over $\mathbb{Z}_N$. Exactly d entries take the high lift $\delta = N{-}d$ and $N{-}d$ take $\delta = -d$, giving $\sum_i \delta(i,d)^2 = d(N{-}d)^2 + (N{-}d)d^2 = Nd(N{-}d)$. Summing: $S_{\mathrm{nc}} = N \sum_{d=1}^{N-1} d(N{-}d) = N^2(N^2{-}1)/6$.

Since $\|A\|_F^2 = (N^2/4)\,S$, the ratio is $S_{\mathrm{c}}/S_{\mathrm{nc}} = (1/3)/(1/6) = 2$. □

F. Anticommutation $\{S, A\} = 0$

Theorem 7 (Anticommutation)**.** *For the* $\mathrm{GL}(2, \mathbb{Z}_N)$ *Qameah with odd prime* N*,*

$$SA + AS = 0. \tag{9}$$

Proof. **(1) Circulant–Hankel decomposition.** A is an antisymmetric circulant (Proposition 4); S is a symmetric Hankel matrix (Sec. II E).

(2) Complementarity. The offset $c = (N{-}1)/2$ gives the identity $f(d) + f(N{-}d) = N{-}1$ for the circulant generating function, which forces the Hankel/circulant cross-terms to satisfy $a(r)\,G(r) + a(N{-}r)\,G(N{-}r) = 0$ by oddness of a under the complementary pairing, where $G(r)$ denotes the Hankel Gram entries.

(3) DFT diagonalization. In the DFT basis, A (circulant) is diagonal with purely imaginary eigenvalues, while S (Hankel) is anti-diagonal. Hence $SA + AS$ is simultaneously diagonal and anti-diagonal, so it vanishes.

(4) Mode-by-mode cancellation. Summing the cross-terms from step (2) over all modes r confirms $SA + AS = 0$ entry-wise.

Verified: $\|SA + AS\|_F = 0$ to machine precision for $N = 3, 5, 7, 11, 13$; verified computationally, code available at `https://github.com/Insightful-Calculations/qhots_v7_calc`. □

Lemma 8 (General Circulant–Hankel Anticommutation)**.** *Theorem 7 holds for* any *antisymmetric circulant paired with* any *symmetric Hankel matrix (Fourier duality: diagonal* $\times$ *anti-diagonal* $= 0$*). The Qameah-specific content is Theorem 9, which requires the eigenvalue ratio* $|\hat{a}_k| = N|\sigma_k|$*. Machine-verified in* `QHOTSLean/SpectralIdentity.lean`*.*

Theorem 9 (Spectral Identity)**.** *For the* $\mathrm{GL}(2,\mathbb{Z}_N)$ *Qameah with odd prime* N*,*

$$A^2 = -N^2\, S^2. \tag{10}$$

Machine-verified in `QHOTSLean/SpectralIdentity.lean`*.*

Proof. The circulant eigenvalues of A are $\lambda_k^{(A)} = i\,\omega_k$ where ω_k are real, and those of S are $\lambda_k^{(S)}$ real. Since S and A share the circulant eigenbasis, we compute in that basis:

$$(A^2)_k = (\lambda_k^{(A)})^2 = -\omega_k^2. \tag{11}$$

The Fourier duality proof gives $|\hat{a}_k| = N|\hat{\sigma}_k|$ algebraically: the DFT diagonalises both A and S simultaneously, and the complementarity identity $f(d) + f(N-d) = N-1$ forces $(\lambda_k^{(S)})^2 = \omega_k^2/N^2$ (BEDROCK #66; proved algebraically via Fourier duality), giving $(A^2)_k = -N^2(\lambda_k^{(S)})^2 = -N^2(S^2)_k$ for all k. Numerical confirmation: $\|A^2 + N^2S^2\|_F = 0$ to machine precision for $N = 3, 5, 7, 11, 13$ (verified computationally). □

G. The Clifford Group Identification

Theorem 10 ($\mathrm{GL}(2,\mathbb{Z}_N)$ as Clifford Gate)**.** *For odd prime* N*, the Qameah gate* $g \in \mathrm{GL}(2,\mathbb{Z}_N)$ *defines a single-quNit Clifford gate via the conjugation action on the Heisenberg–Weyl group:*

$$g \;\mapsto\; U_g \quad \textit{such that} \quad U_g\, X^a Z^b\, U_g^\dagger = X^{a'} Z^{b'}, \tag{12}$$

where $(a', b')^T = g\,(a, b)^T \bmod N$*. This map identifies the quotient* $\mathrm{GL}(2,\mathbb{Z}_N)/Z(\mathrm{GL})$ *with the extended single-quNit Clifford group* $\mathcal{C}_N/\mathrm{HW}(N)$[10]*.*

Proof sketch. The identification follows from three standard facts[1,2]:

(i) The normalizer of $\mathrm{HW}(N)$ in $U(N)$ is the Clifford group $\mathcal{C}_N$. Every Clifford unitary U acts on $\mathrm{HW}(N)$ by conjugation, permuting the Weyl operators: $UW_{a,b}U^\dagger = \omega^{\phi(a,b)}\, W_{a',b'}$ for some phase ϕ and linear map $(a, b) \mapsto (a', b')$.

(ii) Modding out phases, the map $(a, b) \mapsto (a', b')$ must preserve the commutation relation $[W_{a,b}, W_{c,d}] \propto \omega^{ad-bc}$, so the induced map lies in the group of 2×2 matrices over $\mathbb{Z}_N$ preserving the symplectic form $ad - bc$ up to a scalar. For prime N, this group is exactly $\mathrm{GL}(2,\mathbb{Z}_N)$ (since every invertible matrix preserves the symplectic form up to its determinant).

(iii) Conversely, every $g \in \mathrm{GL}(2,\mathbb{Z}_N)$ lifts to a unitary U_g (unique up to phase) implementing the corresponding conjugation. The kernel of the map $U_g \mapsto g$ is $\mathrm{HW}(N)$ itself, giving $\mathcal{C}_N/\mathrm{HW}(N) \cong \mathrm{GL}(2,\mathbb{Z}_N)$.

For the Qameah gate specifically, $\det(g) = s = 2^{-1} \neq 1$, so g lies in the *extended* Clifford group $\mathrm{GL}(2,\mathbb{Z}_N) \setminus \mathrm{SL}(2,\mathbb{Z}_N)$. The subgroup $\mathrm{SL}(2,\mathbb{Z}_N)$ corresponds to the "symplectic" or "restricted" Clifford group that preserves the symplectic form exactly (not just up to scalar). □

H. MUB Partition of Weyl Operators

The Weyl labeling induces a natural partition of the $N^2 - 1$ non-identity operators into *mutually unbiased bases* (MUBs).

Definition 11 (Slope Partition)**.** For each Weyl operator W_{ℓ_1,ℓ_2} with $(\ell_1, \ell_2) \neq (0, 0)$, define the *slope*

$$\sigma(\ell_1, \ell_2) = \begin{cases} \ell_2\, \ell_1^{-1} \bmod N & \text{if } \ell_1 \neq 0, \\ \infty & \text{if } \ell_1 = 0. \end{cases} \tag{13}$$

The $N^2 - 1 = N(N+1) - (N+1) = (N-1)(N+1)$ non-identity operators partition into $N+1$ classes of $N-1$ operators each, indexed by the slopes $\sigma \in \mathbb{Z}_N \cup \{\infty\}$:

$$\mathrm{HW}(N)/Z(\mathrm{HW}) \setminus \{I\} = \bigsqcup_{\sigma \in \mathbb{Z}_N \cup \{\infty\}} \mathcal{M}_\sigma, \qquad |\mathcal{M}_\sigma| = N - 1. \tag{14}$$

Operators within the same class *commute*: if $W_{a,b}$ and $W_{c,d}$ share a slope, then $ad - bc \equiv 0 \pmod N$, so $[W_{a,b}, W_{c,d}] = 0$. Each class $\mathcal{M}_\sigma$ therefore generates an abelian subgroup of $\mathrm{HW}(N)$, whose simultaneous eigenbasis defines a MUB.

For $N = 11$ this gives 12 MUBs of 10 commuting Pauli operators each, partitioning all 120 non-identity Weyl operators. This is the maximal MUB set for an 11-dimensional system, consistent with the general result that prime-dimensional systems admit exactly $N+1$ MUBs[4]. Numerical verification confirms: 12 groups of size 10, all intra-group commutators vanishing, covering all 120 operators with no overlap (verified computationally; code available at `https://github.com/Insightful-Calculations/qhots_v7_calc`).

III. THE PROCA EQUATION AS CASIMIR EIGENVALUE

A. Discrete Differential Forms on K_N

We equip the complete graph K_N with a combinatorial differential complex. Label the N vertices $0, 1, \ldots, N-1$. The three relevant vector spaces are:

- **0-forms** $C^0 \cong \mathbb{R}^N$: scalar functions on vertices.
- **1-forms** $C^1 \cong \mathbb{R}^{\binom{N}{2}}$: antisymmetric functions on oriented edges (i, j) with $i < j$. We identify these with the strictly upper-triangular entries A_{ij} of an antisymmetric $N \times N$ matrix.
- **2-forms** $C^2 \cong \mathbb{R}^{\binom{N}{3}}$: antisymmetric functions on oriented triangles (i, j, k) with $i < j < k$.

The *coboundary operators* are defined by

$$(B_1 f)_{ij} = f_j - f_i, \tag{15}$$
$$(B_2 A)_{ijk} = A_{ij} + A_{jk} + A_{ki}. \tag{16}$$

The composition $B_2 \circ B_1 = 0$ holds identically (a discrete analogue of $d^2 = 0$). The *combinatorial 1-Laplacian* is the up-Laplacian

$$\Delta_1^{\mathrm{up}} = B_2^T B_2, \tag{17}$$

which maps $C^1 \to C^1$. (We omit the down-Laplacian $B_1 B_1^T$ since A will turn out to be orthogonal to its range.)

Definition 12 (Discrete Curvature)**.** For a 1-form $A \in C^1$, the discrete curvature is the 2-form $F = B_2 A$, with components

$$F_{ijk} = A_{ij} + A_{jk} + A_{ki}. \tag{18}$$

The adjoint $\delta F = B_2^T F$ is the discrete codifferential ("divergence" of curvature), mapping 2-forms back to 1-forms.

B. Theorem: Proca Eigenvalue $B_2^T B_2\, A = N\, A$

Theorem 13 (Proca Eigenvalue)**.** *Let B_2 be the coboundary operator on the complete graph K_N for odd prime N, and let A be the antisymmetric part of the deviated Qameah magic square. Then*

$$B_2^T B_2\, A = N\, A. \tag{19}$$

That is, A is an eigenform of the combinatorial 1-Laplacian with eigenvalue N.

Proof. Since A is an antisymmetric circulant (Proposition 4), its eigenvectors are the discrete Fourier modes $\hat{e}_k(j) = \omega^{kj}$ with $\omega = e^{2\pi i/N}$. On the complete graph K_N, the up-Laplacian $B_2^T B_2$ acts diagonally in this basis. For a circulant 1-form with Fourier coefficient $\hat{A}_k$, a standard computation gives

$$(B_2^T B_2\, A)_{ij} = \sum_{m\neq i,j} F_{ijm} = \sum_{m\neq i,j} \big(A_{ij} + A_{jm} + A_{mi}\big). \tag{20}$$

The sum over m of $A_{jm} + A_{mi}$ reduces, by circulant symmetry ($A_{jm} = a(m{-}j)$, $A_{mi} = a(i{-}m)$), to $(N{-}2)\, A_{ij}$ minus A_{ij} itself, since $\sum_{m=0}^{N-1} a(m) = 0$ for an antisymmetric circulant and the two excluded terms contribute $-A_{ij}$ each. The correction from the $(N{-}2)$ remaining vertices contributes $+2\, A_{ij}$. In total:

$$(B_2^T B_2\, A)_{ij} = (N{-}2)\, A_{ij} + 2\, A_{ij} = N\, A_{ij}. \tag{21}$$

Alternatively, the circulant eigenvalues of $B_2^T B_2$ on K_N are $\lambda_k = N$ for all $k \neq 0$ (the standard spectrum of the complete-graph up-Laplacian restricted to the antisymmetric sector[5]), and A, having zero row sums, lies entirely in the $k \neq 0$ subspace. Numerical verification: $\|B_2^T B_2\, A - N\, A\|_F = 0$ to machine precision for all primes $N = 3, 5, 7, 11, 13, 17, 19, 23$ (verified computationally). □

C. Hodge Decomposition: Transverse and Longitudinal

The Hodge decomposition of C^1 on K_N splits into three orthogonal summands:

$$C^1 = \mathrm{im}(B_1) \;\oplus\; \mathcal{H}^1 \;\oplus\; \mathrm{im}(B_2^T), \tag{22}$$

where $\mathcal{H}^1 = \ker(B_2) \cap \ker(B_1^T)$ is the harmonic space. For the complete graph K_N ($N \geq 3$), the first homology vanishes: $\mathcal{H}^1 = 0$. The dimensions are:

$$\dim C^1 = \binom{N}{2}, \tag{23}$$

$$\dim\, \mathrm{im}(B_1) = N - 1 \qquad \text{(gradient/longitudinal)}, \tag{24}$$

$$\dim\, \mathrm{im}(B_2^T) = \binom{N}{2} - (N-1) = \binom{N{-}1}{2} \qquad \text{(co-exact/transverse)}. \tag{25}$$

For $N = 11$: $\binom{11}{2} = 55 = 10 + 45$, decomposing the 55-dimensional edge space into 10 longitudinal (gradient) modes and 45 transverse (co-exact) modes.

Theorem 14 (Transversality)**.** *The antisymmetric part A of the* $\mathrm{GL}(2, \mathbb{Z}_N)$ *magic square is* 100% *co-exact: $A \in \mathrm{im}(B_2^T)$, equivalently $B_1^T A = 0$.*

Proof. The condition $B_1^T A = 0$ states that $(B_1^T A)_i = \sum_j A_{ij} = 0$ for every vertex i. This is the zero row-sum condition. Since A is an antisymmetric circulant, its row sums are $\sum_j a(j{-}i) = \sum_{d=0}^{N-1} a(d)$. But $a(d) + a(N{-}d) = 0$ for all d (antisymmetry of the circulant), and the terms pair completely when N is odd, giving $\sum_d a(d) = 0$. Hence A is co-exact for *any* antisymmetric circulant on K_N—in particular, for any $\mathrm{GL}(2, \mathbb{Z}_N)$ magic square with any odd N. This is a 5-line proof using only circulant antisymmetry, with no reference to the specific generating function $a(d)$. □

The Proca equation (19) describes a massive spin-1 field on K_N with mass squared $m^2 = N$. The co-exactness $B_1^T A = 0$ is the lattice Lorenz condition, automatic for the Proca field.

D. Yang-Mills Action and Lagrangian

Define the discrete Yang-Mills action (curvature energy) and the Proca Lagrangian:

$$S_{\mathrm{YM}} = \sum_{i<j<k} F_{ijk}^2, \tag{26}$$

$$\mathcal{L}_{\mathrm{Proca}} = -\frac{1}{4} S_{\mathrm{YM}} + \frac{N}{2} \sum_{i<j} A_{ij}^2. \tag{27}$$

Proposition 15 (Universal Yang-Mills Action)**.** *For the* $\mathrm{GL}(2, \mathbb{Z}_N)$ *Qameah with odd prime* N*,*

$$S_{\mathrm{YM}} = \frac{N^5(N^2-1)}{24}. \tag{28}$$

Proof. The discrete curvature F_{ijk} takes only two values: $F_{ijk} = N^2$ when all three gaps $(j-i, k-j, i-k)$ mod N are odd, and $F_{ijk} = 0$ otherwise. This follows from the circulant formula: $a(d)/N = (N-d)/2$ for d odd and $a(d)/N = -d/2$ for d even, so the sum $a(d_1) + a(d_2) + a(d_3)$ with $d_1 + d_2 + d_3 \equiv 0 \pmod N$ equals N^2 when all three displacements are odd and vanishes otherwise. The number of all-odd-gap triangles is $N(N^2-1)/24$ (a combinatorial identity for complete graphs with odd N). Each contributes N^4 to S_{YM}, giving (28). □

The on-shell identity (obtained by summing the Proca equation (19) over all edges) is

$$S_{\mathrm{YM}} = N \sum_{i<j} A_{ij}^2, \tag{29}$$

which is the discrete Proca virial relation. Using this, the Lagrangian simplifies:

$$\mathcal{L}_{\mathrm{Proca}} = -\frac{S_{\mathrm{YM}}}{4} + \frac{S_{\mathrm{YM}}}{2} = \frac{S_{\mathrm{YM}}}{4} = \frac{N^5(N^2-1)}{96}. \tag{30}$$

The ratio of kinetic to potential energy is $T/V = -1/2$, the standard Proca virial theorem. Note that $\mathcal{L} \neq 0$: the configuration is *not* Bogomol'nyi–Prasad–Sommerfield (BPS).

The Hessian of $\mathcal{L}_{\mathrm{Proca}}$ has all positive eigenvalues (verified $N = 3, 5, 7, 11, 13$), so A is a local minimum of the Proca action.

E. Curvature Quantization

Proposition 16 (Curvature Quantization)**.** *For the* $\mathrm{GL}(2, \mathbb{Z}_N)$ *Qameah, the discrete curvature takes only two values:*

$$F_{ijk} \in \{0,\ N^2\}. \tag{31}$$

This $\mathbb{Z}_2$-valued curvature parallels lattice gauge theory with gauge group $\mathbb{Z}_2$: every plaquette is either flat ($F = 0$) or maximally curved ($F = N^2$). The flat triangles are those containing at least one even-gap edge; the curved triangles are precisely the $N(N^2-1)/24$ all-odd-gap triangles. For $N = 11$, there are 55 curved triangles out of $\binom{11}{3} = 165$ total, a fraction of exactly 1/3.

F. Solution Space Dimension

The eigenvalue N of the up-Laplacian on K_N has multiplicity $\binom{N-1}{2}$ (the full co-exact subspace). Hence the Proca equation $B_2^T B_2 A = N A$ is satisfied by *every* antisymmetric 1-form with zero row sums, not just the Qameah A.

Proposition 17 (Solution Space)**.** *The solution space of $B_2^T B_2 A = N A$ on K_N has dimension*

$$\dim \ker(B_2^T B_2 - N I)\big|_{C^1} = \binom{N-1}{2} = \dim \mathfrak{so}(N-1) = \dim \Lambda^2 \mathbb{R}^{N-1}. \tag{32}$$

For $N = 11$: $\binom{10}{2} = 45$.

Remark 18 (Honest Constraint). The 45-dimensional degeneracy means the Proca equation alone does *not* uniquely select the Qameah's antisymmetric part A. Additional constraints—integrality of entries, the magic square condition, and membership in the $\mathrm{GL}(2, \mathbb{Z}_N)$ orbit—are required to single out A within this 45-dimensional space. This is analogous to selecting a lattice vector within a real vector space: the lattice structure is essential, not decorative.

IV. BRST CODE $[[N, 1, (N{+}1)/2]]$

A. Nilpotent Operators $L_\pm$

The S/A decomposition and the spectral identity $A^2 = -N^2 S^2$ (Theorem 9, proved algebraically via Fourier duality; BEDROCK #66 of[6]) combine to produce a nilpotent pair.

Definition 19 (BRST Operators)**.** Define the operators

$$L_\pm = S \pm \frac{A}{N}, \tag{33}$$

acting on $\mathbb{R}^N$ (column vectors).

The operators $L_\pm$ act on $\mathbb{R}^N$, and the code parameters $[[N, 1, (N{+}1)/2]]$ are computed over the reals. Promoting this to a quantum error-correcting code over $\mathbb{F}_q$ or $\mathbb{C}^N$ is an open problem (Section VII F).

Theorem 20 (Nilpotency)**.** *For the* $\mathrm{GL}(2, \mathbb{Z}_N)$ *Qameah with odd prime N,*

$$L_+^2 = 0, \qquad L_-^2 = 0. \tag{34}$$

Proof. Expanding L_+^2:

$$L_+^2 = S^2 + \frac{SA + AS}{N} + \frac{A^2}{N^2}. \tag{35}$$

The cross term vanishes by the anticommutation theorem (9): $SA{+}AS = 0$. The remaining terms combine using the spectral identity $A^2 = -N^2 S^2$:

$$L_+^2 = S^2 + \frac{A^2}{N^2} = S^2 + \frac{-N^2 S^2}{N^2} = S^2 - S^2 = 0. \tag{36}$$

The proof for L_-^2 is identical (the cross term changes sign twice: once from $-A/N$ and once from the anticommutator, so the contributions cancel the same way). This proof is purely algebraic and holds for all odd primes N satisfying $\{S, A\} = 0$ and $A^2 = -N^2 S^2$. □

B. Equal Columns and Gauge Structure

Theorem 21 (Complementary Column Degeneracy)**.** *For the* $\mathrm{GL}(2,\mathbb{Z}_N)$ *Qameah with odd* $N \geq 3$*,*

$$(L_+)_{r,\,i} = (L_+)_{r,\,N-1-i} \qquad \textit{for all } r,\, i. \tag{37}$$

That is, column i *and column* $N{-}1{-}i$ *of* L_+ *are identical.*

Proof. The entries of L_+ decompose via $S =$ Hankel, $A =$ circulant:

$$(L_+)_{r,i} = h\big((r{+}i) \bmod N\big) + \frac{a\big((r{-}i) \bmod N\big)}{N}. \tag{38}$$

Under the substitution $i \to N{-}1{-}i$:

$$(L_+)_{r,\,N-1-i} = h\big((r{-}1{-}i) \bmod N\big) + \frac{a\big((r{+}1{+}i) \bmod N\big)}{N}. \tag{39}$$

Applying the identity $h(u) = a(u{+}1)/N$ to both original terms:

$$(L_+)_{r,i} = \frac{a(r{+}i{+}1)}{N} + \frac{a(r{-}i)}{N}, \tag{40}$$

$$(L_+)_{r,\,N-1-i} = \frac{a(r{-}i)}{N} + \frac{a(r{+}i{+}1)}{N}. \tag{41}$$

These are identical. □

The column degeneracy pairs $\lfloor (N{-}1)/2 \rfloor$ columns, so $\mathrm{rank}(L_+) \leq (N{+}1)/2$.

C. Image Structure: Gauge Vectors

The column degeneracy determines the image of L_+ completely.

Proposition 22 (Weight-2 Gauge Vectors)**.** *For each* i *with* $0 \leq i < (N{-}1)/2$*, the column difference*

$$\mathbf{g}_i = \mathbf{e}_i - \mathbf{e}_{N-1-i} \tag{42}$$

lies in $\ker(L_+)$*, where* $\mathbf{e}_i$ *denotes the* i*-th standard basis vector of* $\mathbb{R}^N$*. More precisely,* $L_+\,\mathbf{g}_i = L_+\,\mathbf{e}_i - L_+\,\mathbf{e}_{N-1-i} = 0$ *by Theorem 21.*

Thus $\ker(L_+) \supseteq \mathrm{span}\{\mathbf{g}_0, \ldots, \mathbf{g}_{(N-3)/2}\}$, a subspace of dimension $(N{-}1)/2$. These gauge vectors have *weight* 2 (exactly two nonzero entries, with values ± 1). The remaining kernel element is the all-ones vector $\mathbf{1} = (1, 1, \ldots, 1)^T$, which satisfies $L_+\mathbf{1} = (S + A/N)\,\mathbf{1} = S\,\mathbf{1} + 0 = 0$ (since $S\,\mathbf{1} = 0$ by zero row sums and $A\,\mathbf{1} = 0$ by antisymmetric circulant row sums). So $\dim\ker(L_+) = (N{+}1)/2$.

Meanwhile, $\mathrm{im}(L_+)$ has dimension $\mathrm{rank}(L_+) = N - (N{+}1)/2 = (N{-}1)/2$. Numerical computation shows that $\mathrm{im}(L_+)$ is spanned by precisely the weight-2 gauge vectors $\{\mathbf{g}_0, \ldots, \mathbf{g}_{(N-3)/2}\}$ (verified $N = 3, \ldots, 23$), yielding the self-referential structure $\mathrm{im}(L_+) \subset \ker(L_+)$—as required by nilpotency.

D. $H^1(L_+) = 1$: Unique Vacuum

The first BRST cohomology is the quotient

$$H^1(L_+) = \frac{\ker(L_+)}{\mathrm{im}(L_+)}. \tag{43}$$

Theorem 23 (One-Dimensional Cohomology)**.** *For every odd prime N with $3 \leq N \leq 61$ (verified numerically via SVD rank computation),*

$$\dim H^1(L_+) = 1, \tag{44}$$

generated by the all-ones vector $\mathbf{1}$.

Proof sketch. From the preceding analysis: $\dim\ker(L_+) = (N{+}1)/2$ and $\dim\operatorname{im}(L_+) = (N{-}1)/2$. Since $\operatorname{im}(L_+) \subset \ker(L_+)$, the cohomology dimension is

$$\dim H^1 = \frac{N+1}{2} - \frac{N-1}{2} = 1. \tag{45}$$

The representative is $\mathbf{1}$: it lies in $\ker(L_+)$ (since $S\,\mathbf{1} = A\,\mathbf{1} = 0$) but *not* in $\operatorname{im}(L_+)$ (which consists of vectors with coordinate sum zero, since each generator $\mathbf{g}_i = \mathbf{e}_i - \mathbf{e}_{N-1-i}$ has zero component sum). Hence $[\mathbf{1}]$ is the unique nonzero cohomology class.

Numerical verification: for each prime $N = 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61$, the matrix L_+ has rank $(N{-}1)/2$ and nullity $(N{+}1)/2$, with $\operatorname{im}(L_+)$ of dimension $(N{-}1)/2$, giving $\dim H^1 = 1$ (verified computationally). □

E. Homological Distance

Definition 24 (Homological Distance)**.** The *distance* of the code defined by L_+ is the minimum weight of a vector in $\ker(L_+) \setminus \operatorname{im}(L_+)$:

$$d = \min\big\{\, \mathrm{wt}(\mathbf{v}) : \mathbf{v} \in \ker(L_+),\ \mathbf{v} \notin \operatorname{im}(L_+) \,\big\}, \tag{46}$$

where $\mathrm{wt}(\mathbf{v})$ counts the number of nonzero entries.

Theorem 25 (Homological Distance)**.** *For odd prime N with $3 \leq N \leq 61$,*

$$d = \frac{N+1}{2} = s. \tag{47}$$

Proof sketch. Any $\mathbf{v} \in \ker(L_+) \setminus \operatorname{im}(L_+)$ must have nonzero component sum (since $\operatorname{im}(L_+)$ has zero component sum). Write $\mathbf{v} = \alpha\,\mathbf{1} + \mathbf{w}$ with $\alpha \neq 0$ and $\mathbf{w} \in \operatorname{im}(L_+)$. The image vectors $\mathbf{w}$ are spanned by the weight-2 gauge vectors $\mathbf{g}_i = \mathbf{e}_i - \mathbf{e}_{N-1-i}$. Each $\mathbf{g}_i$ can zero out one coordinate of $\mathbf{1}$ (by choosing $\alpha = 1$ and subtracting the appropriate unit) but simultaneously creates a coordinate of value 2 in the complementary position. The optimal strategy is to zero out $(N{-}1)/2$ coordinates (using $(N{-}1)/2$ independent gauge vectors), leaving the remaining $(N{+}1)/2$ coordinates nonzero. No further reduction is possible because $\operatorname{im}(L_+)$ has dimension $(N{-}1)/2$ and each generator affects exactly two coordinates. Hence $d = N - (N{-}1)/2 = (N{+}1)/2$.

Numerical verification by exhaustive search over $\ker(L_+) \setminus \operatorname{im}(L_+)$ confirms $d = (N{+}1)/2$ for all primes $3 \leq N \leq 61$ (verified computationally). □

F. Code Parameters $[[N, 1, (N{+}1)/2]]$

The code distance d is computed as the minimum Hamming weight of any vector in $\ker(L_+) \setminus \operatorname{im}(L_+)$. Parametrising the one-dimensional coset as $\mathbf{1} + \operatorname{im}(L_+)$, we minimise $\mathrm{wt}(\mathbf{1} + w)$ over $w \in \operatorname{im}(L_+)$ by exhaustive search, yielding $d = (N{+}1)/2$ for all primes $3 \leq N \leq 61$.

Theorem 26 (BRST Code Parameters)**.** *For odd prime N with $3 \leq N \leq 61$ (verified numerically), the BRST cohomology of the nilpotent operator $L_+ = S + A/N$ defines a quantum code with parameters*

$$[[N,\ 1,\ (N{+}1)/2]]. \tag{48}$$

The code encodes $k = 1$ logical quNit in N physical quNits with distance $d = (N{+}1)/2 = s$.

Proof sketch. The three code parameters are established as follows.

(i) **Physical qudits** $n = N$**:** The operator L_+ acts on $\mathbb{R}^N$, with the N coordinates corresponding to the N vertices of K_N.

(ii) **Logical dimension** $k = 1$**:** By Theorem 23, $\dim H^1(L_+) = 1$. The unique logical state is encoded in the cohomology class [**1**].

(iii) **Distance** $d = (N{+}1)/2$**:** By Theorem 25, the minimum-weight nontrivial cohomology representative has weight $(N{+}1)/2$. Any error affecting fewer than $\lfloor (d{-}1)/2 \rfloor = (N{-}1)/4$ physical qudits can be detected; any error affecting fewer than $d/2$ can be corrected.

The proof that $\dim H^1 = 1$ is rigorous given the verified rank computations; the distance proof relies on the exhaustive search over the finite kernel. A fully algebraic proof valid for all primes remains open. □

Proposition 27 (BRST Laplacian)**.** *The BRST Laplacian* $\Delta = L_+L_- + L_-L_+ = 4S^2$.

Proof. From $\{S, A\} = 0$ we have $SA = -AS$, so $[S, A] = 2SA$. From Theorem 9, $A^2 = -N^2S^2$. Then $L_+L_- = S^2 - 2SA/N + S^2 = 2S^2 - 2SA/N$ and $L_-L_+ = 2S^2 + 2SA/N$. Summing: $\Delta = 4S^2$. The spectral gap is $4\min_k s_k^2 = N^2/\sin^2(6\pi/N)$. □

G. Spectral Geometry of the BRST Laplacian

Proposition 27 identifies the BRST Laplacian as $\Delta = 4S^2$, pinning its full spectrum to the eigenvalues of the symmetric matrix S. The results below develop the spectral geometry of Δ: its zeta function and functional determinant (Remark 28), the two-sheeted multiplicity structure, the $N{=}2$ SUSY quantum mechanics realisation (Proposition 30), and the CPT decomposition encoded in the supercharge $Q = S + A/N$.

Remark 28 (Spectral Zeta and Functional Determinant). The nonzero eigenvalues of $\Delta = 4S^2$ are $\lambda_k = 4s_k^2$, where $s_k = N/(2\sin(k\pi/N))$ for $k = 1, \ldots, N-1$. A direct computation using the identity $\prod_{k=1}^{N-1} 2\sin(k\pi/N) = N$ gives the spectral zeta

$$\zeta_\Delta(p) \;=\; \sum_{k=1}^{N-1} \lambda_k^{-p} \;=\; \frac{1}{4^p N^{2p-1}} \binom{2p}{p}, \qquad p \in \mathbb{Z}_{>0}. \tag{49}$$

This is BEDROCK #74 of the QHOTS foundation. The regularised functional determinant then follows by $\zeta'_\Delta(0)$:

$$\det{}'(\Delta) \;=\; N^{2(N-2)} \cdot 2^{2(N-1)} \;=\; \left[N^{N-2} \cdot 2^{N-1}\right]^2, \tag{50}$$

which is the square of the Cayley tree-count formula N^{N-2} (with an extra factor 2^{N-1} from the four-fold scaling $\Delta = 4S^2$).

Remark 29 (Spectral Halving). The two summands L_+L_- and L_-L_+ commute (both equal $2S^2 \pm 2SA/N$ modulo anticommutator terms that cancel in the sum) and each accounts for exactly half of the eigenvalue multiplicity of Δ. Consequently the BRST Laplacian spectrum splits as a two-sheeted cover of the S-spectrum: each eigenvalue $\lambda_k = 4s_k^2$ occurs once in the image of L_+L_- and once in the image of L_-L_+. Note that free convolution of L_+ and L_- is obstructed by nilpotency ($L_\pm^2 = 0$), so the standard free-probability semicircle law does not apply here.

Proposition 30 ($N = 2$ SUSY Quantum Mechanics)**.** *The nilpotent pair* (L_+, L_-) *realises an* $N{=}2$ *supersymmetric quantum mechanics algebra. Setting* $Q = L_+$ *(supercharge) and* $Q^\dagger = L_-$*, the three defining relations*

$$Q^2 = 0, \quad (Q^\dagger)^2 = 0, \quad \{Q, Q^\dagger\} = H, \tag{51}$$

are satisfied with $H = \Delta = 4S^2$ (Proposition 27). Every nonzero eigenvalue of H therefore appears with multiplicity two, corresponding to a boson–fermion pair; the Witten index $W = \mathrm{Tr}[(-1)^F e^{-\beta H}]$ vanishes because the spectrum is purely bosonic at zero energy ($\ker S$ contributes one unpaired ground state) and all excited states are paired. SUSY is unbroken yet trivially so: the single zero-mode gives $W = 0$ without a dynamical symmetry-breaking mechanism.

Proof. $Q^2 = L_+^2 = (S + A/N)^2 = S^2 + \{S, A/N\} + A^2/N^2 = S^2 + 0 - S^2 = 0$, using $\{S, A\} = 0$ and $A^2 = -N^2S^2$. Identically $(Q^\dagger)^2 = 0$. For the anticommutator, $\{Q, Q^\dagger\} = L_+L_- + L_-L_+ = \Delta = 4S^2$ by Proposition 27. The Witten index follows from $\ker H = \ker S$ (one-dimensional) and the paired nonzero spectrum. □

Remark 31 (CPT Bridge via $Q = S + A/N$). The supercharge $Q = S + A/N$ acts simultaneously as the CP operator in eigenvalue space. The two summands carry distinct discrete symmetries: S is persymmetric (Hankel mod N), so it maps eigenvalue index $k \mapsto m+1-k$ (spatial parity P); A is an exact circulant, hence DFT-diagonal, so it flips grade $V_+ \to V_-$ (charge conjugation C). Time reversal acts as $T\colon A \to -A$, $Q \to Q^\dagger$, which is the adjoint involution already present in the SUSY algebra. Together these give a CPT triple

$$C = \text{grade flip}, \quad P = (k \mapsto m + 1 - k), \quad T = (A \to -A),$$

with A/N coupling DFT-paired frequencies $k \leftrightarrow N - k$, which in the folded S-eigenbasis realises the anti-diagonal exchange. The CPT product satisfies $\{CP, T(CP)\} = 4S^2 = H$, so the Hamiltonian is itself the CPT commutator—the spectral chain closes on its own symmetry group.

Proposition 32 (Superalgebra Structure)**.** *The algebra generated by S and A under the relations $\{S, A\} = 0$ and $A^2 = -N^2S^2$ admits a natural $\mathbb{Z}/2$-grading: even elements $\{S^k\}$ (geometry) and odd elements $\{S^kA\}$ (gauge). The product rule $S^aA \cdot S^bA = -N^2S^{a+b+2}$ shows that odd×odd=even with coefficient $-N^2S^2$, making this a superalgebra over $k[S]$ with Clifford-like matrix-valued metric. The gauge field A is algebraically subordinate to the geometry S: every element of the algebra reduces to a polynomial in S times at most one factor of A.*

Proof. Define the even subspace $\mathcal{A}_0 = k[S]$ and the odd subspace $\mathcal{A}_1 = k[S]\cdot A$. The grading is preserved under products: $\mathcal{A}_0 \cdot \mathcal{A}_0 \subset \mathcal{A}_0$ (obvious); $\mathcal{A}_0 \cdot \mathcal{A}_1 \subset \mathcal{A}_1$ (obvious); $\mathcal{A}_1 \cdot \mathcal{A}_1 \subset \mathcal{A}_0$, since $S^aA \cdot S^bA = S^a(AS^b)A = S^a(-S^bA)A = -S^{a+b}A^2 = -S^{a+b}(-N^2S^2) = N^2S^{a+b+2}$, using $\{S, A\} = 0$ to anticommute A past S^b. The direct sum $\mathcal{A} = \mathcal{A}_0 \oplus \mathcal{A}_1$ is therefore a $\mathbb{Z}/2$-graded associative algebra (superalgebra) with metric form $\langle A, A\rangle = A^2 = -N^2S^2$. □

Theorem 33 (Clifford Algebra $\mathrm{Cl}(1,1)$)**.** *The generators A/N and J (the exchange matrix, $J^2 = I$) satisfy the three defining relations of the Clifford algebra $\mathrm{Cl}(1,1)$:*

$$(A/N)^2 = -S^2, \qquad J^2 = I, \qquad \{A/N,\, J\} = 0. \tag{52}$$

Hence $\mathrm{Cl}(1,1) \cong \mathrm{Mat}(2,\mathbb{R})$ is the ambient algebra of the Qameah gate. Three physically distinct structures emerge as faces of this single Clifford object:

- ***BRST supercharge:*** *$Q = S + A/N$ is an odd element; $Q^2 = S^2 - S^2 = 0$ encodes the nilpotency constraint.*
- ***SUSY grading:*** *the decomposition $V = V_+ \oplus V_-$ is induced by the projectors $P_\pm = \frac{1}{2}(I \pm AJ/(NS))$, which span the even sub-algebra $\mathrm{Cl}^0(1,1) \cong \mathbb{R} \oplus \mathbb{R}$.*
- ***CPT symmetry:*** *J is precisely one of the two Clifford generators; conjugation by J acts as CPT on the spectrum of S.*

BRST, SUSY, and CPT are therefore not independent symmetries imposed on the Qameah gate: they are the three canonical aspects of the single $\mathrm{Cl}(1,1)$ structure.

Proof. The relation $(A/N)^2 = -S^2$ is BEDROCK #62 ($\{S, A\} = 0$ implies $A^2 = -N^2 S^2$ up to the normalisation A/N). It remains to establish the two anticommutators involving J.

Lemma (circulant conjugation). Let C be any $N \times N$ circulant matrix and J the exchange (reversal) matrix with $(J)_{ij} = \delta_{i+j,N+1}$. Then $JCJ = C^{\mathsf{T}}$.

Proof. A circulant C has $(C)_{ij} = c_{(j-i) \bmod N}$. Then $(JCJ)_{ij} = (C)_{N+1-i,\, N+1-j} = c_{(i-j) \bmod N} = (C^{\mathsf{T}})_{ij}$.

Corollary 1: $\{J,\, A/N\} = 0$**.** The circulant A is antisymmetric ($A^{\mathsf{T}} = -A$), so $JAJ = A^{\mathsf{T}} = -A$, i.e. $JA = -AJ$, giving $\{J, A\} = 0$ and therefore $\{J,\, A/N\} = 0$.

Corollary 2: $\{J,\, S\} = 0$**.** S is a skew-persymmetric Hankel matrix: $JS = -SJ$ by direct index computation ($(JS)_{ij} = S_{N+1-i,\, j} = -S_{j,\, N+1-i} = -(SJ)_{ij}$, using the skew-persymmetry condition $S_{ij} = -S_{N+1-j,\, N+1-i}$). Hence $\{J, S\} = 0$.

All three anticommutators are therefore algebraic consequences of the index structure of A and S. Verified computationally for all odd $N = 3, \ldots, 37$ (sim1502, 18/18 checks pass). The exchange matrix J is the chirality operator that maps $|s_k\rangle \to |-s_k\rangle$. □

H. Information-Theoretic Structure

The anticommutation $\{S, A\} = 0$ has a sharp information-theoretic consequence: the signal sub-channel (Hankel matrix S) and the noise sub-channel (circulant matrix A) carry *identical* classical capacity for every odd N.

Proposition 34 (Equal Classical Capacity)**.** *Let $\mathcal{N}_S$ and $\mathcal{N}_A$ be the quantum channels whose Kraus operator is $S/\|S\|_F$ and $A/\|A\|_F$, respectively. Then $C(\mathcal{N}_S) = C(\mathcal{N}_A)$ for all odd N.*

Proof sketch. Classical capacity is determined by the Shannon entropy of the row distributions of $|K|^2$ (matrix element squares normalised per row). A direct computation shows that S^2 and A^2 have identical *sorted row multisets*: every row of S^2 is a permutation of the corresponding row of A^2. Hence their channel matrices induce the same row-entropy profile and the same classical capacity. The row-multiset identity follows from $A^2 = -N^2 S^2 + c\, I$ (a consequence of $\{S, A\} = 0$ and the trace condition) after normalisation. □

Remark 35 (Democratic SNR). The relation $\{S, A\} = 0$ forces the signal-to-noise ratio to be *uniform* across all $N - 1$ non-zero spectral modes:

$$\mathrm{SNR}_k \;=\; \frac{s_k^2}{a_k^2} \;=\; \frac{1}{N^2}, \qquad k = 1, \ldots, N-1.$$

No mode is privileged; the vacuum treats all frequencies democratically.

Remark 36 (Unbounded Quantum Advantage). Although the classical capacities are equal, the Holevo capacity χ (accessible with entangled decoding) grows strictly faster than C as N increases. Numerical evaluation gives $\chi/C \approx 1.7$ at $N = 3$, 4.0 at $N = 11$, and 4.8 at $N = 29$, with the ratio growing without bound. The quantum advantage is thus an intrinsic feature of the Qameah geometry that has no finite cap.

V. WHY $N = 11$: FOUR SELECTION CRITERIA

Four independent criteria converge uniquely at $N = 11$.

A. Fibonacci–Triangular Coincidence

The complete graph K_N has $\binom{N}{2}$ edges. For $N = 11$,

$$\binom{11}{2} = 55 = T_{10} = F_{10}, \tag{53}$$

where $T_m = m(m+1)/2$ denotes the m-th triangular number and F_m the m-th Fibonacci number. The equation $T_m = F_m$ (same index on both sides) has only finitely many solutions: $m = 1, 2, 4, 8, 10^7$. Among these, $m = 10$ gives the largest solution $T_{10} = F_{10} = 55$, and $N = m + 1 = 11$ is the *unique prime* for which $\binom{N}{2}$ is simultaneously triangular and Fibonacci with the same index.

B. E_8 Root Count

Define $R(N) = 2(N^2 - 1)$, twice the dimension of the adjoint representation of SU(N). For $N = 11$:

$$R(11) = 2(121 - 1) = 240 = |\Phi(E_8)|, \tag{54}$$

where $\Phi(E_8)$ is the root system of the exceptional Lie algebra E_8. The number 240 counts the shortest vectors in the E_8 root lattice Γ_8 and equals $|\mathrm{Aut}(\Gamma_8)|/|W(D_8)|$ in the McKay correspondence[8]. Among all primes N with $3 \le N \le 61$, the equation $2(N^2 - 1) = 240$ has the unique solution $N = 11$. No smaller exceptional algebra (E_6 with 72 roots, E_7 with 126) arises from a prime N.

C. Perfect Code Distance

The BRST code of Sec. IV F has distance $d = (N+1)/2$. For $N = 11$:

$$d = \frac{11 + 1}{2} = 6 = 1 + 2 + 3, \tag{55}$$

a *perfect number* (equal to the sum of its proper divisors). This is the weakest of the four criteria, included for completeness. Among all odd primes N with $3 \le N \le 61$, the distances are $d = 2, 3, 4, 6, 7, 9, 10, 12, 15, 16, 19, 21, 22, 24, 27, 29, 30, 31$. Of these, $d = 6$ is the *only* perfect number (the next perfect number is 28, which would require $N = 55$, a composite). Hence the perfect-number condition on d selects $N = 11$ uniquely within the tested range, and conditionally for all primes (under the standard conjecture that even perfect numbers have the form $2^{p-1}(2^p - 1)$, the next candidate distance $d = 28$ requires $N = 55 = 5 \times 11$, which is not prime).

D. Euler Characteristic Identity

Define the graph Euler characteristic (vertices − edges + triangles):

$$\chi(K_N) = N - \binom{N}{2} + \binom{N}{3}. \tag{56}$$

Theorem 37 (Euler Characteristic Selection)**.** *Among odd primes N with $3 \le N \le 61$, the identity*

$$\chi(K_N) = N^2 \tag{57}$$

holds if and only if $N = 11$. Furthermore, $|\mathrm{HW}(N)/Z(\mathrm{HW}(N))| = N^2$ for every odd prime N, so the identity reads: the graph Euler characteristic of the carrier graph equals the number of Heisenberg–Weyl cosets.

Proof. Setting $\chi(K_N) = N^2$:

$$N - \frac{N(N-1)}{2} + \frac{N(N-1)(N-2)}{6} = N^2, \tag{58}$$

$$1 - \frac{N-1}{2} + \frac{(N-1)(N-2)}{6} = N, \tag{59}$$

$$6 - 3(N-1) + (N-1)(N-2) = 6N, \tag{60}$$

$$6 - 3N + 3 + N^2 - 3N + 2 = 6N, \tag{61}$$

$$N^2 - 6N + 11 = 6N, \tag{62}$$

$$N^2 - 12N + 11 = 0, \tag{63}$$

$$(N-1)(N-11) = 0. \tag{64}$$

The solutions are $N = 1$ (degenerate) and $N = 11$. Among odd primes, $N = 11$ is the unique solution.

For the Heisenberg–Weyl group $\mathrm{HW}(N)$ over $\mathbb{Z}_N$ (with N prime), $|\mathrm{HW}(N)| = N^3$ and $|Z(\mathrm{HW}(N))| = N$, so $|\mathrm{HW}(N)/Z(\mathrm{HW}(N))| = N^2$—the number of cosets, indexing the N^2 Weyl operators $X^a Z^b$ with $(a, b) \in \mathbb{Z}_N^2$. □

E. Quadruple Coincidence

The four criteria involve Fibonacci theory, Lie algebra root systems, number-theoretic perfectness, and simplicial topology. An additional triple coincidence reinforces the selection:

$$|E_8^+| = |\text{Pauli frame}| = |2I| = 120, \tag{65}$$

where E_8^+ is the positive E_8 roots, the Pauli frame comprises the $N^2 - 1 = 120$ non-identity Weyl operators for $N = 11$, and $2I$ is the binary icosahedral group whose McKay graph is the E_8 Dynkin diagram[8].

F. Predictive verification at $N = 13$

To verify that the spectral identities derived in this paper are universal—not artifacts of the special role of $N = 11$ in the selection criteria above—we test all Complete Spectral Theorem (CST) predictions at $N = 13$. The $\mathrm{GL}(2, \mathbb{Z}_{13})$ magic square was constructed independently; ten scalar properties were predicted from the closed-form CST expressions and then verified by exact computation (rational arithmetic via `sympy` and `numpy` eigenvalues).

TABLE I. CST predictions versus computed values at $N = 13$. All ten identities pass; five representative entries are shown.

	Identity	Predicted	Status
(a)	Magic constant $m_c = N(N^2+1)/2$	1105	✓
(b)	Frobenius weight $W = S_{\mathrm{YM}}/N$	199927	✓
(c)	Yang–Mills action $S_{\mathrm{YM}} = N^5(N^2-1)/24$	2 599 051	✓
(d)	Functional determinant $\prod \mu_k^2 = N^{2(N-1)}$	13^{23}	✓
(e)	Galois group $\mathrm{Gal}(q/\mathbb{Q}) \cong \mathbb{Z}_{(N-1)/2}$	$\mathbb{Z}_6$	✓

All 10/10 predictions pass, including the full set of elementary symmetric polynomials e_k and the csc-formula for oscillator frequencies ω_k (relative error 5.5×10^{-14} after the Part A2 scaling correction $\mu_k = \omega_k \sqrt{N^2/(N^2-1)}$). This confirms the CST as a genuine universal theorem for $\mathrm{GL}(2, \mathbb{Z}_N)$ magic squares.

VI. NEWTON SUM IDENTITIES

The Lehmer roots $r_k = 2\cos(2\pi k/N)$ for $k = 1, \ldots, (N-1)/2$ are the zeros of the N-th Lehmer polynomial $\Phi_N(x)/\Phi_1(x)$, expressed in the variable $x = \omega + \omega^{-1}$ where $\omega = e^{2\pi i/N}$. Their power sums exhibit universal identities depending only on N and the central binomial coefficients.

A. Statement and Proof

Theorem 38 (Newton Sum Identities for Lehmer Roots)**.** *Let N be an odd prime and $r_k = 2\cos(2\pi k/N)$ for $k = 1, \ldots, (N-1)/2$. Define the power sums*

$$P_n = \sum_{k=1}^{(N-1)/2} r_k^n. \tag{66}$$

Then:

(i) ***Odd power sums.*** *Writing $n = 2n' + 1$ for $0 \le n' < (N-1)/2$:*

$$P_{2n'+1} = -4^{n'}. \tag{67}$$

Equivalently, the full-range sum satisfies

$$\sum_{k=1}^{N-1} (2\cos(2\pi k/N))^{2n'+1} = -2^{2n'+1} \qquad (0 \le n' < (N-1)/2). \tag{68}$$

(ii) ***Even power sums.*** *For even $n = 2m$ with $1 \le m < N$:*

$$P_{2m} = \frac{N\binom{2m}{m} - 4^m}{2}. \tag{69}$$

Proof. We use the binomial expansion of $(e^{i\theta} + e^{-i\theta})^n = (2\cos\theta)^n$:

$$(2\cos\theta)^n = \sum_{j=0}^{n} \binom{n}{j} e^{i(n-2j)\theta}. \tag{70}$$

Setting $\theta = 2\pi k/N$ and summing over $k = 1, \ldots, N-1$:

$$\sum_{k=1}^{N-1} r_k^n = \sum_{j=0}^{n} \binom{n}{j} \underbrace{\sum_{k=1}^{N-1} e^{2\pi i k(n-2j)/N}}_{S(n-2j)}. \tag{71}$$

The inner sum $S(m) = \sum_{k=1}^{N-1} e^{2\pi i k m/N}$ satisfies

$$S(m) = \begin{cases} N-1 & \text{if } N \mid m, \\ -1 & \text{if } N \nmid m. \end{cases} \tag{72}$$

Odd case. When n is odd and $1 \le n \le N-2$, the condition $N \mid (n-2j)$ requires $n \equiv 2j \pmod{N}$. Since $0 \le j \le n \le N-2$, the quantity $n - 2j$ ranges over $\{n, n-2, \ldots, -(n-2), -n\}$. For $n < N$, none of these values is a nonzero multiple of N

(since $|n-2j| \le n < N$), and $n-2j=0$ requires $j=n/2$, which is not an integer when n is odd. Hence $S(n-2j)=-1$ for all j, giving

$$\sum_{k=1}^{N-1} r_k^n = -\sum_{j=0}^{n} \binom{n}{j} = -2^n. \tag{73}$$

This gives -2^n for the full-range sum. The half-range sum $P_n = \sum_{k=1}^{(N-1)/2}$ equals exactly half the full-range sum because $r_{N-k} = r_k$ (since $\cos(2\pi(N-k)/N) = \cos(2\pi k/N)$). Thus

$$P_n = \frac{1}{2}\sum_{k=1}^{N-1} r_k^n = -\frac{2^n}{2} = -2^{n-1} \qquad (n \text{ odd},\ 1 \le n \le N-2). \tag{74}$$

For the specific formulation with r_k^{2n+1} (odd exponent $2n+1$), the full-range sum over $k = 0,\ldots,N-1$ (including $k=0$ where $r_0=2$) is $-2^{2n+1}+2^{2n+1}=0$, but the $k=1,\ldots,N-1$ sum is -2^{2n+1} and the half-range sum is $-2^{2n}=-4^n$. Hence:

$$\boxed{\sum_{k=1}^{(N-1)/2} r_k^{2n+1} = -4^n} \qquad (0 \le n < (N-1)/2). \tag{75}$$

Even case. When $n=2m$ is even, the condition $N \mid (2m-2j)$, *i.e.*, $N \mid 2(m-j)$, has solutions. Since N is an odd prime and $\gcd(2,N)=1$, this reduces to $N \mid (m-j)$, giving $j=m$ (and $j=m\pm N$, etc., when in range). For $0 \le j \le 2m$ with $m<N$, only $j=m$ contributes the "resonance" $S(0)=N-1$. All other terms give $S(2m-2j)=-1$. Therefore:

$$\begin{aligned}\sum_{k=1}^{N-1} r_k^{2m} &= \binom{2m}{m}(N-1) + \left(\sum_{j\ne m}\binom{2m}{j}\right)(-1)\\ &= \binom{2m}{m}(N-1) - \left(2^{2m} - \binom{2m}{m}\right)\\ &= N\binom{2m}{m} - 4^m.\end{aligned} \tag{76}$$

Since $r_{N-k}=r_k$, the half-range sum is half the full-range:

$$\boxed{P_{2m} = \frac{N\binom{2m}{m} - 4^m}{2}} \qquad (1 \le m < N). \tag{77}$$

□

Corollary 39 (Range Asymmetry)**.** *The odd formula holds for $0 \le n < (N-1)/2$; the even formula for $1 \le m < N$ (approximately twice the odd range). The asymmetry arises because $N \mid 2(m-j)$ with $\gcd(2,N)=1$ delays the second resonance to $m \ge N$.*

Defining $\zeta_S(n) = P_n/N^n$, the even identity gives $\zeta_S(2m) = \binom{2m}{m}/(2N^{2m-1}) - 4^m/(2N^{2m})$, recovering Catalan-type spectral densities as $N \to \infty$.

B. Lehmer Polynomial at $x=2$

The Lehmer polynomial for prime N is $p_N(x) = \prod_{k=1}^{(N-1)/2}(x-r_k)$, the minimal polynomial of $2\cos(2\pi/N)$ over $\mathbb{Q}$[12]. A classical identity gives:

Proposition 40 (Conductor Identity). *For odd prime N:*

$$p_N(2) = N. \tag{78}$$

Proof. The N-th cyclotomic polynomial $\Phi_N(x) = \sum_{j=0}^{N-1} x^j$ satisfies $\Phi_N(1) = N$. Under the substitution $x = e^{2\pi i/N}$ and the factorization $\Phi_N(x) = \prod_{k=1}^{(N-1)/2}(x-\omega^k)(x-\omega^{-k})$, we obtain $\Phi_N(1) = \prod_{k=1}^{(N-1)/2}|1-\omega^k|^2 = \prod_{k=1}^{(N-1)/2}(2-2\cos(2\pi k/N)) = \prod_{k=1}^{(N-1)/2}(2-r_k) = p_N(2)$. Hence $p_N(2) = \Phi_N(1) = N$. □

The Lehmer polynomial evaluates at $x = 2$ to give N itself.

C. Elementary Symmetric Polynomials and Primality

The Newton power sums P_n determine the elementary symmetric polynomials $e_k = e_k(r_1, \ldots, r_m)$ of the Lehmer roots via Newton's identities, where $m = (N-1)/2$.

Proposition 41 (Unified e_k Formula). *Let N be an odd prime, $m = (N-1)/2$, and $r_k = 2\cos(2\pi k/N)$ for $k = 1, \ldots, m$. For $k = 1, \ldots, m$:*

$$e_k = N^{2k}\,\frac{\binom{m+k}{2k}}{2k+1}. \tag{79}$$

The rational factor $B(m,k) := \binom{m+k}{2k}/(2k+1)$ is the ballot number $B_{m,k}$ (also a Catalan-type quotient), and it is an integer for every $k < m$. The formula has been verified numerically for all odd primes $N \in \{5, 7, 11, 13, 17\}$.

Remark 42 (Primality Criterion). $B(m,k)$ is an integer for all $k = 1, \ldots, m-1$ if and only if $N = 2m+1$ is prime. The proof uses Kummer's theorem: the q-valuation of $\binom{m+k}{2k}/(2k+1)$ counts carries in base-q addition; for $q \mid N$ prime, all carries vanish. Composite N produces a non-integer $B(m,k)$ for at least one k, providing a number-theoretic companion to the algebraic structure of the Lehmer polynomial.

Remark 43 (Characteristic Polynomial as Hypergeometric Series). The Lehmer polynomial $p_N(x) = \prod_{k=1}^{m}(x - r_k)$ can be expressed as:

$$p_N(x) = x^N \cdot {}_2F_1\left(-m,\, m{+}1;\, \tfrac{3}{2};\, \frac{N^2}{4x^2}\right). \tag{80}$$

This identifies the Lehmer polynomial as a Gegenbauer-type hypergeometric function, consistent with the $e_k = N^{2k}B(m,k)$ structure where N^{2k} arises from the argument $N^2/(4x^2)$.

VII. DISCUSSION

A. One Axiom

Table I summarizes all properties derived from $s = 2^{-1} \bmod N$.

Once N is an odd prime, $s = 2^{-1}$ determines everything; the four selection criteria of Sec. V then fix $N = 11$.

B. Connections to Quantum Error Correction

The BRST code shares its homological structure with CSS codes[4,9,11]: L_+ plays the role of a parity-check matrix, $\ker(L_+)$ is the code space, and $\mathrm{im}(L_+)$ defines stabilizer

TABLE II. Twenty properties derived from $s = 2^{-1} \bmod N$.

Property	Reference	From s via
$\mathrm{GL}(2, \mathbb{Z}_N)$ gate entries	Def. 1	$g = \left[\begin{smallmatrix} s & s-1 \\ s & s \end{smallmatrix}\right]$
Weyl labeling	Def. 3	$g \bmod N$
S/A decomposition	Sec. II E	$h(u) = a(u+1)/N$
Anticommutation $\{S, A\} = 0$	Thm. 7	$f(d) + f(N-d) = 0$
Clifford isomorphism	Thm. 10	$g \in \mathrm{GL}(2, \mathbb{Z}_N)$
Proca eigenvalue N	Thm. 13	$B_2^T B_2 A = NA$
Curvature quantization	Prop. 16	$F \in \{0, N^2\}$
BRST nilpotency	Thm. 20	$L_\pm = S \pm A/N$
Column degeneracy	Thm. 21	$h(u) = a(u+1)/N$
$H^1 = 1$	Thm. 23	rank structure
Distance $d = s$	Thm. 25	$d = (N+1)/2$
Code $[[N, 1, s]]$	Thm. 26	all above
Fibonacci–triangular	Sec. V A	$N = 11$
E_8 root count	Sec. V B	$2(N^2-1)$
Perfect distance	Sec. V C	$d = 6$
$\chi = N^2$	Thm. 37	$(N-1)(N-11) = 0$
Newton odd sums	Thm. 38(i)	roots of unity
Newton even sums	Thm. 38(ii)	roots of unity
Conductor $p(2) = N$	Prop. 40	$\Phi_N(1)$
Spectral zeta	Sec. VI	even sums

equivalence. The distance $d = s$ corrects up to $\lfloor (s-1)/2 \rfloor$ errors; the weight-2 gauge vectors $\mathbf{g}_i$ are unusually sparse stabilizer generators. Whether $[[11, 1, 6]]$ admits efficient syndrome extraction or achieves a useful threshold remains open.

Proposition 44 (Transversal Clifford Action)**.** *The* $[[N, 1, (N+1)/2]]$ *BRST code is a maximum-distance-separable (MDS) stabilizer code over the Heisenberg–Weyl group* HW_N*. By the Rains*[14] *and Ashikhmin–Knill*[15] *framework for MDS stabilizer codes, the full Clifford group* $\mathrm{Cl}(N)$ *acts transversally on the code: every single-qudit Clifford gate can be implemented fault-tolerantly by applying the corresponding* HW_N*-coset operation to each of the* N *physical qudits independently. The 121 Qameah cells label the 121 cosets of* $Z(\mathrm{HW}_{11})$ *in* HW_{11}*, providing a complete syndrome alphabet consistent with transversal Clifford action. This is compatible with the Eastin–Knill theorem:* $\mathrm{Cl}(N)$ *is transversal but not universal, so no contradiction arises.*

Proof sketch. An $[[N, 1, d]]$ stabilizer code with $d = (N+1)/2$ saturates the quantum Singleton bound $d \leq n - k + 1 = N$, confirming MDS status. Theorem 5 of Rains[14] establishes that any MDS stabilizer code over the Heisenberg–Weyl group admits a transversal Clifford group action; Ashikhmin–Knill[15] extend this to odd prime alphabets. The syndrome completeness follows from $|\mathrm{HW}_N / Z(\mathrm{HW}_N)| = N^2 = 121$ for $N = 11$ (Theorem 37), matching the number of Qameah cells. □

C. Conference Matrix Conjecture

Remark 45 (Paley Conference Matrix: Honest Negative). The sign pattern of A follows even/odd parity of the index difference, not the Legendre symbol. The Paley conference identification is therefore **false** for the Qameah (verified computationally for $N = 3, 5, 7, 11, 13$; code available at `https://github.com/Insightful-Calculations/qhots_`

`v7_calc`). The Paley matrix has known spectrum in terms of $\sqrt{N}^3$, but this connection does not apply to the Qameah's antisymmetric part. No claim in this paper depends on the Paley identification.

D. Connections to E_8 Lattice Theory

For $N = 11$ the Proca solution space has dimension $\binom{10}{2} = 45 = \dim \mathfrak{so}(10)$. The Qameah A is a distinguished integral element of this space. The coincidences $2(N^2 - 1) = 240 = |\Phi(E_8)|$ and $|E_8^+| = |2I| = 120$ suggest the Qameah sits within E_8 lattice geometry, but an explicit embedding remains open.

E. Triple Selection of s

The value $s = (N+1)/2$ is uniquely selected by three independent constraints: (1) algebraic—$\{S, A\} = 0$ requires circulant A (Theorem 7); (2) variational—s maximizes $\|A\|_F^2$ by a factor of 2 (Proposition 6); (3) number-theoretic—curvature quantisation $F \in \{0, N^2\}$ requires the circulant structure.

As a curiosity, the Gaussian prime $z = 5 + 6i \in \mathbb{Z}[i]$ encodes the key parameters: $N = \mathrm{Re}(z) + \mathrm{Im}(z) = 11$, $d = \mathrm{Im}(z) = 6$, and the magic constant $M = N|z|^2 = 671$. The norm $|z|^2 = 61$ is itself prime, consistent with z being a Gaussian prime, but we do not claim this encoding is more than a mnemonic.

Remark 46 (Klein Bottle Identification). The S/A decomposition admits a topological interpretation as a Klein bottle identification pattern on the index space $\mathbb{Z}_N \times \mathbb{Z}_N$. The symmetric part S (Hankel, invariant under transposition) provides a same-direction identification, while the antisymmetric part A (circulant, sign-reversing under transposition) provides an opposite-direction identification. Same-direction plus opposite-direction is the combinatorial definition of a Klein bottle. The operators $L_\pm = S \pm A/N$, each nilpotent with $L_\pm^2 = 0$, correspond to the two Möbius strips obtained by cutting the Klein bottle along its median. This identification is formalised in the Lean 4 module `KleinBottle.lean`.

F. Open Problems

(1) **Sign pattern structure:** The sign pattern of A follows index-parity, not the Legendre symbol (Remark 45). Understanding the spectral consequences of this parity structure remains open.

(2) **Algebraic distance proof:** $d = (N+1)/2$ is verified for $3 \leq N \leq 61$; a proof for all primes requires showing the gauge vectors are the only minimal-weight image elements.

(3) **Lean4 formalization:** Theorems 38, 37, and 20 are algebraic and formalizable.

(4) **Extremality of A:** Is the Qameah A extremal in the 45-dimensional Proca eigenspace (*e.g.*, lattice norm, packing bound)?

(5) **Physical utility of $[[11, 1, 6]]$:** Threshold, decoding complexity, and comparison with known qudit codes remain open.

(6) **Quantum promotion:** The BRST code is defined over $\mathbb{R}^N$; promoting it to a quantum code over $\mathbb{F}_q$ or $\mathbb{C}^N$ requires establishing suitable stabilizer structure. Proposition 44 shows the MDS property implies transversal Clifford action once this promotion is established.

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Appendix A: Numerical Verification Tables

Table II summarizes the BRST code parameters verified by direct computation; code available at `https://github.com/Insightful-Calculations/qhots_v7_calc`.

TABLE III. BRST code parameters for odd primes $3 \leq N \leq 61$. All primes satisfy $\dim H^1 = 1$ and $d = (N+1)/2$.

N	$\mathrm{rank}(L_+)$	$\dim\ker$	$\dim\mathrm{im}$	$\dim H^1$	d
3	1	2	1	1	2
5	2	3	2	1	3
7	3	4	3	1	4
11	5	6	5	1	6
13	6	7	6	1	7
17	8	9	8	1	9
19	9	10	9	1	10
23	11	12	11	1	12
29	14	15	14	1	15
31	15	16	15	1	16
37	18	19	18	1	19
41	20	21	20	1	21
43	21	22	21	1	22
47	23	24	23	1	24
53	26	27	26	1	27
59	29	30	29	1	30
61	30	31	30	1	31

Table III verifies the Newton sum identities for $N = 11$ (representative case; all primes $3 \leq N \leq 61$ verified).

Table IV verifies the Proca eigenvalue and related invariants.

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TABLE IV. Newton sum verification for $N = 11$. P_n^{num} is the numerical sum; P_n^{thy} is the theoretical prediction.

n	P_n^{num}	P_n^{thy}	Formula
1	-1	-1	$-4^0 = -1$
2	9	9	$(11 \cdot 2 - 4)/2$
3	-4	-4	-4^1
4	25	25	$(11 \cdot 6 - 16)/2$
5	-16	-16	-4^2
6	78	78	$(11 \cdot 20 - 64)/2$
7	-64	-64	-4^3
8	257	257	$(11 \cdot 70 - 256)/2$
9	-256	-256	-4^4
10	874	874	$(11 \cdot 252 - 1024)/2$

TABLE V. Proca eigenvalue verification. λ is the smallest nonzero eigenvalue of $B_2^T B_2$; $\dim\ker$ is the co-exact multiplicity.

N	$\binom{N}{2}$	λ	$\dim\ker(B_2^T B_2 - NI)$	$\binom{N-1}{2}$
3	3	3	1	1
5	10	5	6	6
7	21	7	15	15
11	55	11	45	45
13	78	13	66	66

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